

Session 13: Post Class tests

1. Alfred Inc. is a publicly traded sporting goods company. The company has \$ 250 million in book value of debt, reported interest expenses of \$ 12.5 million in the most recent year and has an average maturity of 5 years for the debt. The pre-tax cost of debt for the firm is currently 4%. What is your best estimate of the market value of debt outstanding at the firm? (You can assume annual interest payments and a marginal tax rate of 40%)
 - a. \$261.12 million
 - b. \$250.00 million
 - c. \$280.29 million
 - d. \$243.01 million
 - e. \$156.68 million
2. Jack's Stores is a retail firm with no conventional debt. It does have operating lease commitments of \$ 12 million each year for the next 8 years. Jack's Stores pre-tax cost of debt is 5%, its cost of capital is 9% and the marginal tax rate is 40%. What is the debt value of operating leases? (The risk free rate is 3%)
 - a. \$96.00 million
 - b. \$84.24 million
 - c. \$77.56 million
 - d. \$66.42 million
 - e. None of the above
3. Faraday Enterprises is a publicly traded company. It currently has 10 million shares trading at \$12/share and \$150 million in book value of equity. The firm also has book value of debt of \$ 75 million and market value of debt of \$ 80 million. The cost of equity for the company is 9%, the pre-tax cost of debt is 4% and the marginal tax rate is 40%. What is the cost of capital?
 - a. 7.4%
 - b. 7.0%
 - c. 7.7%
 - d. 6.36%
 - e. None of the above
4. Lester Inc. has 5 million shares outstanding, trading at \$20/share. The company has one convertible bond, with a face value of \$ 100 million, a ten-year maturity and a coupon rate of 2%; the bond has a market value of \$120 million. If the current cost of equity for the firm is 10% and the pre-tax cost of debt is 5%, what is the cost of capital for the firm? (The marginal tax rate is 40%)
 - a. 5.20%
 - b. 6.18%
 - c. 7.55%
 - d. 8.25%
 - e. None of the above
5. JG Enterprises has 50 million shares, trading at \$4 a share; the cost of equity is 12%. It has debt with a market value of \$ 100 million and a pre-tax cost of debt of 6%. Finally the company has \$100 million in market value of preferred stock; the preferred shares are trading at \$80 a share, with an annual preferred

dividend of \$6/share. If the marginal tax rate is 40%, estimate the cost of capital for the firm.

- a. 8.78%
- b. 8.03%
- c. 9.38%
- d. 9.90%
- e. None of the above

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1. **a. \$261.12 million.** To compute the market value of the debt, discount the expected interest expenses and the principal on the debt at the pre-tax cost of debt
 - $\text{Market value of debt} = 12.5 * (1 - 1.04^{-5}) / 0.04 + 250 / 1.04^5 = \261.12 m
 - The first term is the present value of \$12.5 million as an annuity for 5 years, discounted back at 4%. The second term is the present value of the face value of the debt at the end of year 5.
2. **c. \$77.56 million.** The debt value of leases is the present value of the operating leases at the pre-tax cost of debt.
 - $\text{Debt value of leases} = \text{PV of annuity of \$12 million @5\% for 8 years} = \77.56 million
3. **d. 6.6%.** The first step is to compute the market value weights of debt and equity
 - $\text{Debt to capital ratio} = 80 / (120 + 80) = 40\%$
 - $\text{Cost of capital} = 9\%(.6) + 4\% (1-.4) (.4) = 6.36\%$
4. **c. 7.55%.** The first step is to decompose the convertible bond into its debt and equity components. To do this, value the convertible bond as if it were a straight bond by discounting the coupons and face value back at the pre-tax cost of debt:
 - $\text{Value of straight bond portion} = \$2 \text{ million (PV of annuity for 10 years @5\%)} + \$100 \text{ million} / 1.05^{10} = \76.83 million
 - $\text{Value of conversion option} = \text{Market value of convertible} - \text{Straight bond value} = \$120 - \$76.83 = \43.17 million
 - $\text{Overall value of equity} = \143.17 million
 - $\text{Cost of capital} = 10\% (143.17 / (143.17 + 76.83)) + 5\% (1-.4) (76.83 / (143.17 + 76.83)) = 7.55\%$
5. **a. 8.78%.** First, compute the preferred dividend yield, which is also the cost of preferred stock:
 - $\text{Preferred dividend yield} = \$6 / 80 = 7.5\%$
 - $\text{Cost of capital} = 12\% (200 / 400) + 6\% (1-.4) (100 / 400) + 7.5\% (100 / 400) = 8.78\%$